



Smoothing Methods - Moving Averages Transcript

Hello everyone, welcome to lesson 1 video 2. In this video, we will discuss smoothing methods using moving averages. These methods help reduce short-term fluctuations in data, making the underlying trend easier to see. The main topics in this video include moving averages and different other methods which are used in time series.

In the last lecture, we studied about decomposing the entire time series into its constituents or its components. Now in this, we are going to move ahead into how do we smooth out a time series data. The very basic method that we use for that is called as moving averages.

Now we can have edges in the time series or let's say irregularities in time series which are generally unlikely to be predicted well. They are random fluctuations which can actually can cause unpredictability or bring in unpredictability in the entire system. So the idea of smoothing is to remove those edges, to remove those random fluctuations from the entire analysis so that the entire curve becomes smoothened out.

So the simplest and the most common smoothing technique is called as moving average. And moving average works on the principle of averaging out the time series components of the past and then getting to the newest value. So we average out the values and then get a new value by taking the average of two or three or four components and so on.

You can average out any number of components and then you can get a new value. Now because there are there could be some values which are really large, some values could be really small. If you take average it would be somewhere in between and that's how it removes the fluctuations in the data.

Now if you think of it the entire process that I told you there are there could be some values which are really large. There could be some values which are really small. And if you average them together the moving average would become I mean it would remove the fluctuations from the data.

So a three point moving average it could be what is your past value one step prior. What is your current value. What is your one step after value let's say for example if you're concentrating on month number one what sorry month number two.

What is the value at month one. What is the value of month two and what is the value of month three. In that case the moving average at month two would be these three summations of these three values divided by three.

So average of all of them right. And this way while doing all of these things we will be able to cancel out all the sharp edges that we have in the entire time series. This is a quick demonstration of what I have been speaking about right.

so if you see there are sharp edges in the entire time series here. Now with a time series moving average value of three or let's say if we use three values to smooth out



the time series in this particular case we are taking the first three values. Hence you have a middle value here right.

The next value would be the composition of next three values and so on. So this is how this entire red curve is constituted. The grey curve is the original curve and red curve is what has been constituted using moving averages.

Now closely see how does this entire moving average now this entire moving average curve is it becomes more and more smooth or it smoothens out when you add more and more terms to it. So K is a number of term let's say. So let's slide it if you see here right.

If you add more terms and more terms you see that the entire curve smooths out right. If you add less terms it is actually having certain spikes but as you add more and more terms then the entire curve smooths out and this is what I just wanted to explain you because visually it looks really good when you are trying to interpret how does moving averages works out. What are the pros and cons of moving averages.

The very first thing it is very easy to understand Galgrid that's what you already saw right. It is effectively smooths out the entire random noise which is what you saw during the demonstration I was showing and then it can adapt it to handle seasonality. So let's say you have fluctuations because of seasonality then moving average would be able to remove those as well.

Because if you take a 12 month moving average then you are basically working on one single data point and you are removing the entire seasonal effect from the entire component. The disadvantages are it loses data points as the Bing because you are averaging out you are losing information when you are summing three or four components or five components together but actually getting replacing those four five data points with a single moving average data right. And the second thing is it lags behind the trend when you actually have a trend but generally you do not get the trend while you average out everything but you also lose a lot of information.

Then it gives equal weight to all the data points because you are averaging based on summing of all the data points and dividing by total number of data points. You are doing simple averaging and not weighted averaging. You are not providing weights to some specific points.

Hence there are crucial information which might be missed in the entire analysis using moving averages. I hope this entire series or this entire lecture was important for you to understand how moving averages smoothens out the entire curve and it also brings to your notion about averaging cancels out how does averaging cancels out the peaks or those edges in the time series. And that's what we saw in this in the simple demonstration.



When you start with lesser number of components we have certain edges which is lesser than what you have in the original curve. But as you keep on adding terms the entire curve becomes more and more flat right which is what is the entire essence of moving average. I hope you have got the understanding well.

In summary moving averages are a simple but powerful tool for smoothing out noise and revealing underlying trends in the data. In the next video we will look into how time series analysis can be decomposed into two main components.

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